

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

|                             |                                                 |
|-----------------------------|-------------------------------------------------|
| <b>Product Description:</b> | <b>Zinc oxide</b>                               |
| <b>Cat No. :</b>            | <b>S36702</b>                                   |
| <b>Synonyms</b>             | Chinese white; Zinc white; C.I. Pigment White 4 |
| <b>CAS No</b>               | 1314-13-2                                       |
| <b>Molecular Formula</b>    | O Zn                                            |

**Relevant identified uses of the substance or mixture and uses advised against**

|                             |                          |
|-----------------------------|--------------------------|
| <b>Recommended Use</b>      | Laboratory chemicals.    |
| <b>Uses advised against</b> | No Information available |

**Details of the supplier of the safety data sheet**

|                                          |                                          |
|------------------------------------------|------------------------------------------|
| <b>Importer</b>                          | <b>Supplier</b>                          |
| Fisher Scientific Korea                  | Thermo Fisher Scientific Chemicals, Inc. |
| D5,D6, Incheon Airport Logistics Complex | 30 Bond Street                           |
| 150, Gonghangdong-Ro 296 Beon-Gil        | Ward Hill, MA 01835-8099                 |
| Jung-Gu, Incheon                         |                                          |
| Tel: +82-1661-9555                       |                                          |
| Fax: +82-2-2023-0603                     |                                          |

|                       |                          |
|-----------------------|--------------------------|
| <b>E-mail address</b> | Chem.KR@thermofisher.com |
|-----------------------|--------------------------|

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

Based on available data, the classification criteria are not met

**Health hazards**

Based on available data, the classification criteria are not met

**Environmental hazards**

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

**Label Elements**

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**Signal Word**

**Danger**

**Hazard Statements**

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements**

**Prevention**

P273 - Avoid release to the environment

**Response**

P391 - Collect spillage

**Disposal**

P501 - Dispose of contents/container to industrial incineration plant

**Other Hazards**

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

**NFPA**

**Health**  
1

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

**3.1. Substances**

| Component  | Common Name                                     | CAS No    | Index No | Weight % |
|------------|-------------------------------------------------|-----------|----------|----------|
| Zinc oxide | Chinese white; Zinc white; C.I. Pigment White 4 | 1314-13-2 | KE-35565 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

**Description of first aid measures**

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention if symptoms occur.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.

**Ingestion**

Do NOT induce vomiting. Get medical attention if symptoms occur.

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention if symptoms occur.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

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## Most important symptoms and effects, both acute and delayed

No information available.

## Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

ALFAAS36702

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Keep containers tightly closed in a dry, cool and well-ventilated place.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component  | CAS No    | Korea                                                                              | ACGIH TLV                                              | OSHA PEL                                                                                                                                                                   |
|------------|-----------|------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zinc oxide | 1314-13-2 | STEL: 10 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | (Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) STEL: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup> |

| Component  | CAS No    | European Union | The United Kingdom | Germany                                                                                                                                                        |
|------------|-----------|----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zinc oxide | 1314-13-2 | Not listed     | Not listed         | TWA: 0.1 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.4 mg/m <sup>3</sup><br>Höhepunkt: 4 mg/m <sup>3</sup> |

### ACGIH - Biological Exposure Indices

| Component  | CAS No    | ACGIH - Biological Exposure Indices |
|------------|-----------|-------------------------------------|
| Zinc oxide | 1314-13-2 | Not listed                          |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Personal protective equipment

#### Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water

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system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Off-white Powder Solid

Odor Odorless

Odor Threshold No data available

pH 7 50 g/l aq.sol.(susp)

Melting Point/Range 1975 °C / 3587 °F

Softening Point No data available

Boiling Point/Range No information available

Flash Point No information available Method - No information available

Evaporation Rate Not applicable Solid

Flammability (solid,gas) No information available

Explosion Limits No data available

Vapor Pressure No information available

Vapor Density Not applicable Solid

Specific Gravity / Density 5.600

Bulk Density No data available

Water Solubility 1.6 mg/L (29°C)

Solubility in other solvents No information available

### Partition Coefficient (n-octanol/water)

| Component  | CAS No    | log Pow           |
|------------|-----------|-------------------|
| Zinc oxide | 1314-13-2 | No data available |

Autoignition Temperature No data available

Decomposition Temperature No data available

Viscosity Not applicable Solid

Explosive Properties No information available

Oxidizing Properties No information available

Molecular Formula O Zn

Molecular Weight 81.38

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

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**Hazardous Polymerization** No information available.  
**Hazardous Reactions** No information available.

**Conditions to Avoid**  
Avoid dust formation. Incompatible products.

**Incompatible Materials**  
Strong acids.

**Hazardous Decomposition Products**  
None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

**Inhalation** May cause irritation of respiratory tract. May be harmful if inhaled. Avoid breathing dust or spray mist.  
**Ingestion** May be harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.  
**Eyes** Avoid contact with eyes.  
**Skin** May cause irritation. May be harmful in contact with skin. Avoid contact with skin.

#### Information on Health Hazards

##### (a) acute toxicity;

**Oral** Based on available data, the classification criteria are not met  
**Dermal** No data available  
**Inhalation** No data available

| Component  | CAS No    | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation           |
|------------|-----------|---------------------------|------------------------------|---------------------------|
| Zinc oxide | 1314-13-2 | LD50 > 5000 mg/kg ( Rat ) | LD50 > 2000 mg/kg, 24h (Rat) | LC50 > 5.7 mg/L, 4h (Rat) |

**(b) skin corrosion/irritation;** No data available  
**Test species** rabbit  
**Observational endpoint** No skin irritation

**(c) serious eye damage/irritation;** No data available  
**Test method** Test method B.5  
OECD 405  
**Test species** rabbit  
**Observation end point** No eye irritation

**(d) respiratory or skin sensitization;**  
**Respiratory** No data available  
**Skin** No data available

| Component  | CAS No    | Test method                                              | Test species | Study result    |
|------------|-----------|----------------------------------------------------------|--------------|-----------------|
| Zinc oxide | 1314-13-2 | in vivo<br>OECD Test Guideline<br>406<br>Test method B.6 | guinea pig   | non-sensitising |

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(e) germ cell mutagenicity; No data available

| Component  | CAS No    | Test method                                                            | Test species         | Study result |
|------------|-----------|------------------------------------------------------------------------|----------------------|--------------|
| Zinc oxide | 1314-13-2 | in vitro<br>OECD Test Guideline 471<br>Bacterial Reverse Mutation Test | in vitro: Bacteria   | negative     |
|            |           | in vivo<br>OECD Test Guideline 474<br>Mammalian                        | in vivo<br>Mammalian | negative     |

Mutagenic effects have occurred in experimental animals

(f) carcinogenicity; No data available

| Component  | CAS No    | Test method       | Test species / Duration | Study result      |
|------------|-----------|-------------------|-------------------------|-------------------|
| Zinc oxide | 1314-13-2 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component  | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|------------|-----------|------------|------------|------------|------------|------------|
| Zinc oxide | 1314-13-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; No data available

| Component  | CAS No    | Test method       | Test species / Duration | Study result      |
|------------|-----------|-------------------|-------------------------|-------------------|
| Zinc oxide | 1314-13-2 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects  
No information available.

| Component  | CAS No    | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|------------|-----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component  | CAS No    | Freshwater Fish    | Water Flea        | Freshwater Algae  | Microtox          |
|------------|-----------|--------------------|-------------------|-------------------|-------------------|
| Zinc oxide | 1314-13-2 | LC50: = 1.55 mg/L, | No data available | No data available | No data available |

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|  |  |                          |  |  |  |
|--|--|--------------------------|--|--|--|
|  |  | 96h static (Danio rerio) |  |  |  |
|--|--|--------------------------|--|--|--|

## Persistence and degradability

### **Persistence**

Soluble in water, Persistence is unlikely, based on information available.

### **Degradability**

Not relevant for inorganic substances.

### **Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

## Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Ozone Depletion Potential

| Component  | CAS No    | Ozone Depletion Potential |
|------------|-----------|---------------------------|
| Zinc oxide | 1314-13-2 | Not listed                |

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

#### **UN-No**

UN3077

#### **Proper Shipping Name**

Environmentally hazardous substances, solid, n.o.s.

#### **Technical Shipping Name**

Zinc oxide

#### **Hazard Class**

9

#### **Packing Group**

III

### IATA

#### **UN-No**

UN3077

#### **Proper Shipping Name**

ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.\*

#### **Technical Shipping Name**

Zinc oxide

#### **Hazard Class**

9

#### **Packing Group**

III

### IMDG/IMO

#### **UN-No**

UN3077

#### **Proper Shipping Name**

Environmentally hazardous substances, solid, n.o.s.

#### **Technical Shipping Name**

Zinc oxide

#### **Hazard Class**

9

#### **Packing Group**

III

#### **Marine Pollutant**

Dangerous for the environment

Product is a marine pollutant according to the criteria set by IMDG/IMO

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Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

### International Inventories

| Component  | CAS No    | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|------------|-----------|----------|------|-----------|-------|-----|------|-------|------|------|------|
| Zinc oxide | 1314-13-2 | KE-35565 | X    | 215-222-5 | X     | X   | -    | X     | X    | X    | X    |

| Component  | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y23                      |

| Component  | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|------------|-----------|----------|------------------------------|---------------------------|
| Zinc oxide | 1314-13-2 | Listed   | Not applicable               | Not applicable            |

### Korean National Regulations

| Component  | CAS No    | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|------------|-----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Zinc oxide | 1314-13-2 | Annex 1 - KE-35565                                                  | Not applicable       | Listed                                      |

| Component  | CAS No    | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|------------|-----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Zinc oxide | 1314-13-2 | 2022-1-1099 (>25%)                     | Not applicable                              | Not applicable                                  |

| Component  | CAS No    | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|------------|-----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component  | CAS No    | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|------------|-----------|-------------------|------------------------------------|-----------------------------------------------------------|
| Zinc oxide | 1314-13-2 | > 25% (CCA)       | Not applicable                     | Not applicable                                            |

| Component  | CAS No    | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|------------|-----------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Zinc oxide | 1314-13-2 | Listed                                                       | Not applicable               | Not applicable                         |

| Component  | CAS No    | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|------------|-----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Zinc oxide | 1314-13-2 | Listed                               | Listed                                             | Not applicable                     |

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| Component  | CAS No    | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals                                     | ISHA - Special management materials |
|------------|-----------|-------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                                              | STEL: 10 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | Not applicable                      |

**National Fire Association - Dangerous Substances** Minimum quantity requiring a permit

| Component  | CAS No    | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|------------|-----------|----------------------------|---------------------------|----------------------------------------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
| Zinc oxide | 1314-13-2 | Not applicable             | Not applicable            | Not applicable                                                                   | Not applicable              | Not applicable                     | Not applicable              |

## Control Parameters

| Component  | CAS No    | Korea                                                                              | ACGIH - Biological Exposure Indices |
|------------|-----------|------------------------------------------------------------------------------------|-------------------------------------|
| Zinc oxide | 1314-13-2 | STEL: 10 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component  | CAS No    | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|------------|-----------|----------------------------------|----------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                   | Not applicable             |

**CERCLA** Not applicable

| Component  | CAS No    | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|------------|-----------|-------------------------------------------|--------------------------|-------------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                            | Not applicable           | 1.0 %                         |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Warning.

H410 - Very toxic to aquatic life with long lasting effects.

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

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**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**POW** - Partition coefficient Octanol:Water

**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical incident response training.

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department        |
| <b>Creation Date</b>    | 19-Jan-2010                                        |
| <b>Revision Date</b>    | 12-Jun-2024                                        |
| <b>Revision Number</b>  | 4                                                  |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**